

刘鹏

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研究兴趣

医学图像分析, 深度学习, 电镜图像分析, 全细胞器分割, 半监督学习

教育背景

- | | | |
|-------------|--|--------|
| 2019 – 2024 | 上海交通大学
生物医学工程博士 | 上海, 中国 |
| | <ul style="list-style-type: none">• 导师: 郑国焱教授• 论文: 基于一致性正则化的半监督医学图像分析 | |
| 2015 – 2018 | 中国科学院大学
信号与信息处理硕士 | 北京, 中国 |
| | <ul style="list-style-type: none">• 论文: 基于深度学习的高分辨率遥感图像目标检测研究 | |
| 2011 – 2015 | 中国地质大学
地理信息系统学士 | 北京, 中国 |

工作经历

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|-------------------------|---|-------------|
| 2024 年 8 月 – 至今 | 波士顿学院 (Boston College) | 博士后研究员. |
| | <ul style="list-style-type: none">• 合作导师: 魏东来教授• 从事电镜 (EM) 图像分割研究, 重点关注线粒体分割、细胞核分割和全细胞器分割。 | |
| 2017 年 7 月 – 2019 年 7 月 | 美团集团 | 搜索与推荐算法工程师. |
| | <ul style="list-style-type: none">• 参与旅游业务排序优化, 提升点击率 (CTR) 和转化率 (CVR)。• 相关算法: Graph Embedding, DNN, Wide&Deep | |

发表论文 (* 代表共同一作)

期刊文章

- 1 **P. Liu** and G. Zheng, "Handling imbalanced data: Uncertainty-guided virtual adversarial training with batch nuclear-norm optimization for semi-supervised medical image classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2983–2994, 2022, (JCR Q1).

- 2 **P. Liu** and G. Zheng, "Cvcl: Context-aware voxel-wise contrastive learning for label-efficient multi-organ segmentation," *Computers in Biology and Medicine*, vol. 160, p. 106 995, 2023, (JCR Q1).
- 3 **P. Liu**, M. Zhang, X. Gao, B. Li, and G. Zheng, "Joint lymphoma lesion segmentation and prognosis prediction from baseline fdg-pet images via multitask convolutional neural networks," *IEEE Access*, vol. 10, pp. 81 612–81 623, 2022, (JCR Q2).
- 4 C. Chen*, **P. Liu***, and Y. Song, "Diagnostic performance for severity grading of hip osteoarthritis and osteonecrosis of femoral head on radiographs: Deep learning model vs. board-certified orthopaedic surgeons," *Osteoarthritis Imaging*, vol. 3, no. 2, p. 100 092, 2023.
- 5 Z. Li, K. Chen, **P. Liu**, X. Chen, and G. Zheng, "Entropy and distance maps-guided segmentation of articular cartilage: Data from the osteoarthritis initiative," *International Journal of Computer Assisted Radiology and Surgery*, pp. 1–8, 2022, (JCR Q2).

会议论文

- 1 **P. Liu** and G. Zheng, "Semi-supervised learning regularized by adversarial perturbation and diversity maximization," in *Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021*, Springer, 2021, pp. 199–208.
- 2 **P. Liu** and G. Zheng, "Context-aware voxel-wise contrastive learning for label efficient multi-organ segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 653–662.
- 3 Q. Zhou*, **P. Liu***, and G. Zheng, "Context-aware cutmix is all you need for universal organ and cancer segmentation," in *Fast, Low-Resource, and Accurate Organ and Pan-Cancer Segmentation in Abdomen CT*, Springer, 2023.
- 4 Q. Zhou, **P. Liu**, and G. Zheng, "Partially supervised multi-organ segmentation via affinity-aware consistency learning and cross site feature alignment," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 671–680.

在审论文

- 1 **P. Liu**, B. Shen, L. Liu, Q. Wang, S. Zhang, A. Bhardwaj, I. Arganda-Carreras, K. Narayan, and D. Wei, "Mitoem 2.0: A benchmark for challenging 3d mitochondria instance segmentation from em images," Under Review at Scientific Data, 2025, (JCR Q1).
- 2 **P. Liu** and G. Zheng, "Cr-gloco: Cross-resolution learning via global-local context consistency for semi-supervised 3d medical segmentation," Minor Revision at Computerized Medical Imaging and Graphics, 2025, (JCR Q1).

在研/拟投

- 1 **P. Liu** and D. Wei, *Skelaffinity-seg: Skeleton-guided geometry and global affinity learning for complex 3d tubular instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

- 2 P. Liu and D. Wei, *Mitoverse: A foundation model for mitochondrial morphology, topology, and instance segmentation*, To be submitted to IEEE Transactions on Medical Imaging, 2026.
- 3 P. Liu and D. Wei, *Mixture-of-experts for multi-resolution pan-organelle segmentation in 3d volume electron microscopy*, To be submitted to IEEE Transactions on Medical Imaging, 2026.

竞赛与奖学金

竞赛

- 2023 第二名, MICCAI FLARE 2023 挑战赛. MICCAI 2023 Workshop.
- 2018 排名: 20/120, 美团 MDD Cup 2018. 于 Kaggle MDD Cup 2018.
- 2017 排名: 7/148, 美团 MDD Cup 2017. 于 Kaggle MDD Cup 2017.

奖学金

- 2014 三等奖学金. 中国地质大学.
- 2013 三等奖学金. 中国地质大学.
- 2012 国家励志奖学金. 中国教育部.
二等奖学金. 中国地质大学.

助教经历

- CS 290I: 程序设计思想与方法 (C) 2022 年秋季.
2021 年秋季.
- CS 170: 程序设计思想与方法 (C) 2020 年秋季.
2019 年秋季.

审稿

- IEEE Transactions on Medical Imaging
- Medical Image Analysis
- IEEE Journal of Biomedical and Health Informatics
- Neural Networks
- Computerized Medical Imaging and Graphics
- MICCAI 2022
- MICCAI 2021